The Effect of Spaced Versus Massed Practice on Word Recall Among University Students

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PSY101: Introduction to Psychology, Section 26

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September 7, 2026

Introduction

When preparing for academic evaluation, students need to determine not only the amount of time they need to spend studying, but also how to allocate that time. Many university students prefer to focus their preparation on one long study period just before an exam. This can yield satisfactory short-term performance, but pupils often say that they do not remember it later. The effects of study time distribution on retention, therefore, have practical significance and are a topic of systematic investigation in psychology for more than 100 years.

The spacing effect is explored in the present study. Myers and DeWall (2021) describe it as the phenomenon that distributed study or practice leads to superior long-term retention than massed study or practice. Breaking up a limited amount of study time into a series of discrete periods tends to yield better retention than having a single block of the same amount of study time.

Theoretical Background

First documented systematically by Ebbinghaus (1885/1913), who did a series of experiments on nonsense syllables, which led him to formulate the forgetting curve and the observation that distributed repetitions result in more lasting learning. Cepeda et al. (2006) found a consistent advantage in distributed compared to massed practice and this advantage was greater as the retention interval lengthened in a quantitative synthesis of several hundred verbal recall experiments. Later, Cepeda et al. (2008) proved that there is an optimum spacing interval and it is proportional to the desired retention period.

This impact extends beyond word lists in the laboratory. Baddeley and Longman (1978) were able to show that although postal workers received the same amount of total training time, those who received the training in a distributed manner retained learning better than those who received the training in a massed manner. In a nine-year longitudinal study, Bahrick et al. (1993) found that spaced practice yielded significantly more recall years later, and in their study of 10 learning strategies, assigned to them the highest utility rating, Dunlosky et al. (2013) found that distributed practice also received a high utility rating.

There are two major theories for why spacing helps learning to be retained. According to the encoding variability theory, the state of the mind and the context of each study episode are somewhat different, thus leading to the formation of different memory traces and to a wider range of retrieval cues during distributed study. The study phase retrieval theory suggests that the re-reading of the material is not sufficient to strengthen memory for the earlier information – it needs to be actively retrieved and not merely re-reading – in order for it to be recalled after a

delay. The latter relates the spacing effect to the testing effect cited by Roediger and Karpicke (2006), who showed that retrieval practice is more effective in delayed retention than restudying.

Relevance and Purpose

Student study behaviour is still based on the principle of concentrated preparation, but the effect of spacing is well known. This study aimed to determine if the effect could be replicated with a university student sample in a controlled experiment, and to interpret data in terms of the above theoretical accounts. The study did not depend on the participant's own account of his study schedule or his own description of the study material but it was manipulated directly and the recall measured objectively.

Research Question

Does distributing study time into separate spaced sessions produce higher word recall than studying the same material in one continuous massed session, among university students?

Hypotheses

- **H_0 (null hypothesis):** There is no significant difference in the number of words recalled between participants in the massed condition and participants in the spaced condition.
- **H_a (alternative hypothesis):** Participants in the spaced condition will recall significantly more words than participants in the massed condition. The alternative hypothesis was stated directionally, on the basis of the consistent direction of effect reported in prior literature.

Objectives

The aims of this study were:

- To determine if there is a significant effect of the distribution of study time on the free recall performance.
- To compare the mean scores of recall between a massed-practice and a spaced-practice condition, with equal overall duration of study.
- To perform an appropriate statistical test to test the stated hypothesis.
- To analyze the results in terms of the spacing effect and to existing encoding-retrieval theories of memory.

- To find the methodological weaknesses and to draw practical conclusions for student study behavior and future investigations.

Methodology

Rationale for an Experimental Design

This study did not use a correlational survey or interview design, it was an experimental design. The research question is testing a cause and effect hypothesis, as it is asking if there is a change in recall with study schedule, not just an association. A correlational design would only show a relationship between two variables and would be open to the possibility of a third variable. An experiment allows for causal inferences because the researcher controls the value of the independent variable. There were three additional factors that led to this decision. The study schedule can be directly applied to the participants rather than observed as it happens naturally. An experimental design was used so that both the amount of study time, amount of study materials, amount of retention interval, and test duration could be kept the same, allowing for the distribution of practice to be the only difference. Lastly, the dependent variable might have been measured objectively on a fixed key, rather than by self-report on a subject as it would have done on a study habits survey.

Design

This was an experimental design with a between-subjects design and two levels of the independent variable: study schedule (massed and spaced), and number of words correctly recalled (dependent variable). All subjects were tested in only one condition. The between-subjects design was used instead of the within-subjects design to avoid the contamination of the word list between the two conditions. The ideal experiment has three features that are satisfied by the present design.

- **Manipulation:** The independent variable was actively manipulated by the researchers. The study schedule for each participant was assigned by the research team, irrespective of any characteristic of the participant, either as a single two-minute block or as three separate 40-second blocks.
- **Control group:** The massed condition served as a standard to compare the performance of the spaced condition. If the comparison had not been made, then there would be no inference about the effect of the manipulation.

- **Random assignment:** All participants were equally likely to be assigned to either condition. Allocation was random and uninfluenced by participant attributes, participant preference, or researcher judgement.

Participants

Thirty-six University students (n = 18 in each condition) aged between 22 and 26 years participated. A total of 26 male and 10 female students, from the second, third and fourth undergraduate years, were included in the sample. Convenience sampling was used to recruit participants, and they participated voluntarily without remuneration. This is recognised as a limitation as it used an English language word list, so participants needed to have a good level of English reading and writing.

Random Assignment Procedure

Data collection involved individually administered interview sessions with research team members. All participants were randomly allocated to either the massed or spaced condition at the time of interview. The assignment was randomly determined before the participant started the task and was not affected by year of study, gender, remembered memory ability, memory preference, or the judgment of the interviewer. The way that pre-existing individual differences are evenly distributed across conditions is random assignment. Other factors potentially related to recall such as baseline memory capacity, motivation, familiarity with English vocabulary and prior study habits were not assessed. These are not expected to be highly concentrated in any one group because they were randomly assigned and thus cannot be used to provide a systematic alternative explanation for any observed difference. Two groups were formed of equal size by random assignment.

Materials

The materials consisted of a list of 20 moderately familiar concrete nouns in English, followed by an unrelated filler task, simple arithmetic problems, both between spaced study intervals and during the retention interval; blank recall sheets; a written informed consent form; a brief demographic sheet; and a stopwatch for the standardisation of time across sessions. Words were chosen to avoid semantic and phonological similarity and to minimise clustering on recall.

Operational Definitions

The constructs under investigation were operationally defined as follows.

- **Massed practice:** study of the 20-word list in one continuous, uninterrupted two-minute block.
- **Spaced practice:** study of the same list across three separate 40-second blocks, each separated by approximately one minute of unrelated filler activity. Total study exposure was held equal to the massed condition.
- **Recall:** the number of words, out of 20, correctly reproduced during a two-minute free-recall test. Minor spelling variations were scored as correct where the intended word was unambiguous, following a rule agreed before scoring commenced.
- **Retention interval:** a 10- to 15-minute unrelated filler task between the final study exposure and the recall test, used to prevent continued rehearsal.

Procedure

Each session was about 20-25 minutes long. The informed consent form was read and signed by the participants at the beginning and they were informed that the study would be about memory, without stating the specific hypothesis at this point. The participants were then randomly allocated to a condition. The group that studied the list massed completed a single 2-minute block of silent study of the list, while the group that studied the list spaced completed three blocks of 40 seconds each separated by about a minute of filler activity. A 10- to 15-minute filler task was then given to both groups as a retention interval. The participants were then allowed two minutes to list as many words as they could in any order that they could remember. The recall sheets were marked using a specific key and a complete debrief was given informing each participant of the real aim of the study.

Ethical Considerations

The study was conducted with complete voluntary participation and participants were told that they were free to withdraw at any time without any repercussions. Informed consent was obtained for all participants. Data were collected with identification numbers instead of names and the data are only presented as totals; anonymity was maintained. The hypothesis was not presented to the participants until after the instances of behaviour were recorded so as to avoid demand characteristics. This is considered to be minor non-disclosure, in the sense that no information was withheld at any point and each participant was given a complete description of the aim of the study at the end of the recall test. There was no physical or serious psychological risk beyond engaging in a mild memory exercise.

Data Analysis

The descriptive statistics were obtained for each condition. An independent samples test was used to compare the mean recall of the two conditions in order to test the hypothesis. Effect size Cohen's d was computed and a 95% confidence interval for the difference between means was obtained. The assumptions of normality and homogeneity of variance were checked prior to the main analysis.

Results

Preliminary Assumption Checks

The test determined that recall scores were normally distributed in both the massed condition ($W = .94$, $p = .339$) and spaced condition ($W = .95$, $p = .431$). Results of the Levene test suggested homogeneity of variance ($F = 0.08$, $p = .784$) and the two group standard deviations were very similar (2.07 and 2.06). The parametric test was hence appropriate and no transformation or non-parametric alternative was necessary.

Descriptive Statistics

Descriptive statistics for both conditions are shown in Table 1. Mean number of words recalled was 8.17 words (SD = 2.07) with a range from 4 to 11 words in the Massed condition. The mean number of words recalled was 12.61 (SD = 2.06) for the spaced condition (range 9 – 17). There was a difference of 4.44 words between the mean of the two conditions, corresponding to a 54.4% relative improvement with spaced practice. The means of these conditions are presented in Figure 1 with error bars indicating the standard error for each condition and the spread between conditions is significantly greater than the standard error around each mean.

Table 1

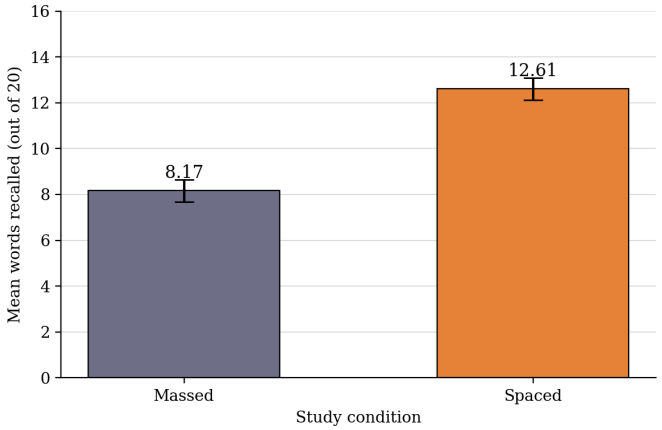
Descriptive Statistics for Recall Scores by Study Condition

Condition	n	M	SD	Mdn	Minimum	Maximum
Massed	18	8.17	2.07	8.50	4	11
Spaced	18	12.61	2.06	13.00	9	17

Note. Recall scores represent the number of words correctly recalled out of a possible 20.

Figure 1

Mean Words Recalled by Study Condition



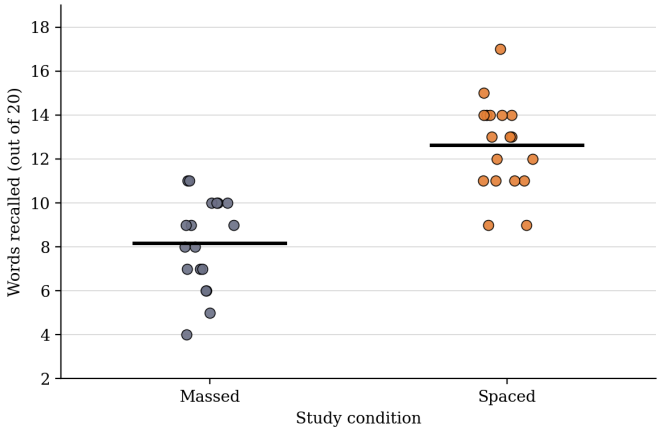
Note. Error bars represent ± 1 standard error of the mean. Maximum possible score = 20.

Distribution of Individual Scores

The information available in the individual scores is not captured in the group means. For each participant, their scores by condition are plotted in Figure 2. There is only a slight overlap between the distributions. In the spaced condition, 18 of 18 reached scores above the average for the massed condition, and 12 of 18 reached scores higher than the highest scores anywhere in the massed condition. The same pattern is shown in a frequency distribution as indicated in table 2. This means that it was not a few high-achieving people that drove the advantage of spacing; it was distributed throughout the condition.

Figure 2

Individual Recall Scores by Study Condition



Note. Each point represents one participant. Horizontal bars indicate condition means.

Table 2

Frequency Distribution of Recall Scores by Study Condition

Score range	Massed (n)	Spaced (n)
4–6	4	0
7–9	8	2
10–12	6	8
13–15	0	7
16–18	0	1

Note. Score ranges are inclusive. Total n = 18 per condition.

Hypothesis Test

The independent-samples t test was used to test the mean recall between conditions. The spaced condition recalled significantly more words than the continuous condition. The 95% confidence interval for the difference from 3.05 words to 5.84 words, which does not contain zero and is entirely in the positive direction, so this interval is not compatible with sampling variation and is compatible with a real advantage of spaced practice. The effect size was large with Cohen's d equal to 2.15, which was significantly higher than the 0.80 criterion for a large effect size.

Summary and Interpretation of Results

The results overall confirm the alternative hypothesis and offer justification for the rejection of the null hypothesis. The direction of the difference is as predicted; the magnitude is substantial not marginal; and the effect was not just in subset of participants, but seen in most, as demonstrated by the lack of overlap in Figure 2. More importantly, the benefit was achieved without an increase in total time spent studying: both conditions were given the same 2 minutes of exposure. Such a difference in observed time is rather due to the distribution of that time than to the amount of time, and this is the main empirical result of the study.

Discussion

It was hypothesized that the division of a given study period would have an effect on free recall. Three separate blocks of 20 words were recalled much better than one continuous block of the same total duration of words. There was no rejection of the null hypothesis and the alternative was accepted. The implications of this outcome for psychological theory, for practical application, and potential limitations in its interpretation are considered in the following sections.

Alignment With the Spacing Effect

The results are consistent with the spacing effect found in previous studies, such as the example provided by Myers and DeWall (2021) and the quantitative evidence provided by Cepeda et al. (2006). The direction of the effect is what was predicted and the effect was observed in a small sample of students in conditions similar to those in the classroom, which is consistent with the reported strength of the effect. In this respect the study is a successful conceptual replication, one of the hundreds of experiments that came before it that demonstrated a finding.

One thing needs to be looked at in greater detail. However, Cepeda et al. (2006) found that the benefits of distributed practice increases with increasing retention interval, and that for very short retention intervals, massed practice can sometimes yield performance at time 1 that is as good as distributed practice. A similar pattern emerged for Roediger and Karpicke (2006), who repeated study and found it to be superior on immediate tests, but inferior on delayed tests. Current research has been conducted over a very short time frame of 10 to 15 minutes and yet still resulted in a significant spacing advantage. This does not disagree with previous work, but rather indicates that the advantage of distribution is experienced quickly and not over a prolonged period as used here.

Theoretical Interpretation

One possible explanation is the encoding variability theory. For each of the 3 spaced blocks, the state of the internal and contextual factors varied slightly, with an arithmetic task between each block. The spaced condition participants therefore encoded the list under three somewhat different sets of conditions, creating a broader set of memory “traces” and “cues.” Those in the Massed condition, which would have been the case if the entire list has been encoded in one continuous state, would have had fewer retrieval pathways and a more narrow trace.

This design can also be explained by study-phase retrieval theory, which is considered to be more powerful. In the spaced condition, both participants would inevitably have seen some items again after the filler activity, thus needing to actively retrieve the items, rather than simply re-read them. This is directly related to the finding by Roediger and Karpicke (2006) that retrieval by itself increases memory retention over the retention of memory from additional exposure. This was not the case in the massed condition as there was no need for retrieval: the material was held continuously in working memory and could be maintained by a simple maintenance rehearsal. A third, related explanation, the deficient-processing explanation, suggests that the more repetitions are presented in a short span, the less conscious they are because they are familiar moment-by-moment. This is because the present data do not discriminate between these accounts

as all three predict the observed direction of effect, but it does show that if any of the mechanisms is correct, it is not due to an additional exposure time.

The Significance of Equal Study Time

This is interesting in theory because both conditions had an equal amount of study exposure, which was 2 minutes. This removes the obvious other explanation, which is just that the spaced group took longer to interact with the material. The difference observed cannot be due to the duration of exposure, as this was the same for both treatments. This aligns with the theory supported by Dunlosky, et al. (2013) that distributed practice leads to better results without any need to spend more total time learning, and named this as one of the most helpful strategies to students to help them learn.

Real-World Implications

The implications of the practice are direct. If the difference in retention is substantial when students have to spend the same amount of time on the same amount of material but the retention is better when the time is spent in more concentrated intervals, then students have a way of enhancing academic results that does not require them to invest more time into the tasks. It would be better for a student who spends 6 hours preparing for an examination to spend those 6 hours over several days than in one day.

The consequences are not limited to the personal level. Incorporating explicit instructions for distributed practice into academic support services, especially since the evidence shows that students inaccurately evaluate their own strategies. When considered from the standpoint of subjective fluency, it seems that this is not a reliable indicator of learning, as Roediger and Karpicke (2006) found that re-reading was more effective than retrieval practice, despite the objective data indicating otherwise. Instructors may also review key information throughout the semester instead of once and for all, as it was shown to Bahrick et al. (1993) that it can impact retention over years and not minutes.

Limitations

There are some restrictions and limitations that should be taken into account in interpreting these findings.

- Sampling. Convenience sampling was used with a single university population to recruit participants. The sample is not representative of the university students as a whole and conclusions should be drawn with caution.
- Number and type of samples. The actual sample size taken was less than the target number of 40-50. It was also imbalanced in terms of gender (26 male, 10 female) further reducing representativeness.
- Retention interval. The study only tested retention within 10-15 minutes, so no evidence can be provided about retention over the days or weeks that are typical of a genuine academic preparation.
- Ecological validity. A list of unrelated concrete nouns is not structurally similar to the material that students are learning. It should not be assumed that the effect measured here would be directly applicable to authentic academic learning.
- Unmeasured variables. There was no assessment of English proficiency, baseline memory ability, motivation or previous study habits. Random assignment controls these as systematic confounds but it does not allow the effects of these to be studied.

Unexpected Challenges

There were some practical problems during the data collection. The slow recruitment rate was the main reason that achieved sample was lower than targeted as individual sessions were planned around participants' class timetables. Second, there were some instances of continued rehearsal during the filler task, which would have jeopardized the retention period; so the filler period had to be carefully supervised, and this was added to the standard procedure once it had been discovered. The third minor problem was the marking of answers that had a few spelling mistakes; a uniform convention was agreed before marking and applied throughout.

Possible Improvements and Future Research

There are several enhancements that could be made to a future replication. The most important would be the use of a much longer retention time, 24 hours or more, so as to be able to study longer-term retention, rather than short-term performance, and to design the study to match conditions under which the spacing effect is theoretically predicted to be greatest. Increasing the sample size, and obtaining a more gender balanced sample using more probability sampling methods would enhance both the statistical power and generalisability.

To enhance ecological validity, the word list could be replaced by meaningful academic content, and to determine if the effect persists, it could test whether the effect persists when the

material has internal structure. If an additional extension is available to make, then the spacing interval could be systematically changed, following the work of Cepeda et al. (2008), to find an optimal spacing interval for a particular retention need. Last, measuring their own self-experienced learning would be useful to see if students are accurate in sensing the value of spacing, which is of practical relevance because of the endurance of massed preparation.

Conclusion

This study aimed to analyze the effect of the distribution of study time on word recall of University students. A manipulation of IV, a comparison condition, and random assignment were used in a between-subjects true experiment with 36 participants learning a 20-word list in either one continuous 2-minute block or three separated 40-second blocks (equal total time). The difference between the two conditions was 4.44 words, with the spaced condition having a higher number of words recalled (M = 12.61) than the massed condition (M = 8.17). The null hypothesis was rejected and the alternative hypothesis supported.

The key to the advantage is that it did not require additional study time. Both conditions received the same exposure as the difference is due to the temporal organization of study and not the quantity of exposure. The result is in line with the spacing effect described by Myers and DeWall (2021) and can be explained with encoding variability and study-phase retrieval models. The effect was consistent across participants, suggesting that it was general rather than specific to certain people. The bottom line for the students is simple: Study time may be better spent moved around rather than extended, and it doesn't require extra effort, it's just a scheduling change.

Firstly, the population sampled has some convenience sampling characteristics, and the sample size is relatively small, most significantly, the study has a short retention time, so that conclusions may be drawn for immediate retention but not for long-term retention. Future work should be extended by a more extended retention period, by using meaningful academic material instead of single words, and systematically by varying the spacing interval. A second potentially fruitful avenue of inquiry would be the relationship of the objective benefit of spacing to student's own awareness of it: the fact that massed preparation did survive the century of contrary evidence to it may mean that the discrepancy between the two is the more significant issue.

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